Relation Algebra

§10. Freyd's Power allegories

(10a)

A **power allegory** is a division allegory with one unary operation on arrows, ∋ **epsiloff**, subject to

1	$\ni_R \square = R\square, \quad \ni_R = \ni_R \square$	
2	$\mathbb{1} \sqsubseteq (R/\ni_R)(\ni_R/R)$	$\ni_R$ is **thick**
3	$\frac{\ni_R}{\ni_R} = \mathbb{1}$	$\ni_R$ is **straight**

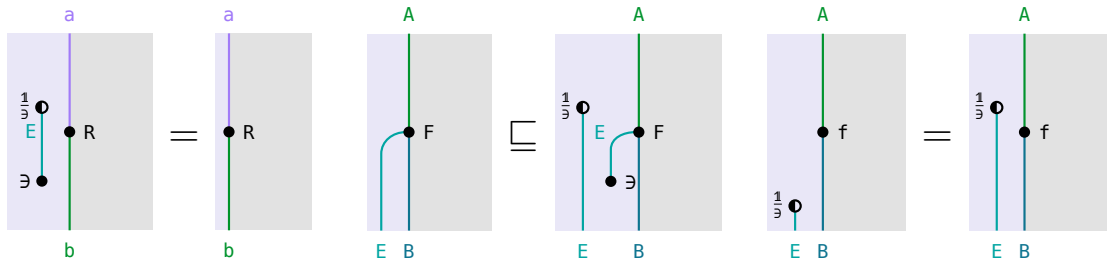
R□ is R's target, an identity arrow. For `R : A→B` write `∋ : PB→B`, dropping the subscript.

`∈≜∋° : A→PA`

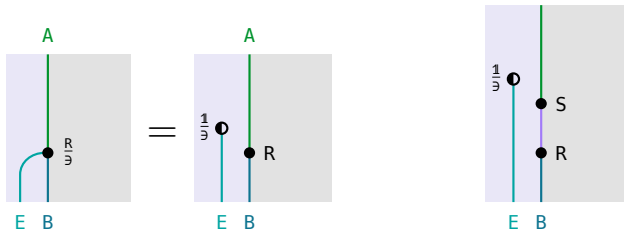
(10b)

	law	the reading
1	$\ni_R \square = R\square, \quad \ni_R = \ni_R \square$	One ∋ per object, not per arrow.
2	∋ is **thick**	**Comprehension**: every x has a set of exactly the people x admires.
3	$\frac{R}{\ni}$ `: A→PB`, for `R : A→B`	convert a relation to a function. `a` $\frac{R}{\ni}$ `={b|a R b}`
4	`Map(`$\frac{R}{\ni}$`)`	
5	$\frac{R}{\ni}$ `∋ = R`	reading the chosen set back through ∋ returns the relation.
6	`Simple(F) ⇒ F ⊑` $\frac{F\ \ni}{\ni}$	A partial choice of sets is inside the total one.
7	the **singleton map** is monic: $\frac{\mathbb{1}}{\ni}$ `(`$\frac{\mathbb{1}}{\ni}$`)° ⊑ 𝟙` two points with the same one-person set are the same point.	The one-person set.
8	$\frac{\ni}{\ni}$ `= 𝟙`	Make the set of a set, then read it back one level down.
9	**fusion:** `Map(f) ⇒` $\frac{f\ R}{\ni}$ `=` $\frac{f}{\ni}$ `R`	Naturality of the unit, `Map(f) ⇒ f` $\frac{\mathbb{1}}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$ `E(f) .` renaming a point and then taking its one-person set is taking the set and renaming inside it.
10	`Map(f); 𝟙□ = ∋□ ⇒` $\frac{f}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$ `f`	Rename first or take singletons first — the fusion row above at R=𝟙.
11	`R□ = ∋□ ⇒ R /ₛ S =` $\frac{R}{\ni}$ `(`$\frac{S}{\ni}$`)°`	x and y match when R sends x and S sends y to the same set.
12	`E(R) ≜` $\frac{\ni\ R}{\ni}$	`E(R): PA→PB`, image of a set of A
13	$\frac{R}{\ni}$ `=` $\frac{\mathbb{1}}{\ni}$ `E(R)`	$\frac{\mathbb{1}}{\ni}$`: x↦{x}`
14	$\frac{S}{\ni}$ `E(R) =` $\frac{S\ R}{\ni}$	absorption — the monad's composition law, $\frac{S}{\ni} \diamond \frac{R}{\ni} = \frac{SR}{\ni}$, §Section 10.3.
15	`E(R) = E(`$\frac{R}{\ni}$`) union`	the image of a set under R is the union of the R-sets of its points.
16	`⊇ ≜ ∋ / ∋`	`xs⊇ys⇔∀a. ys∋a→xs∋a`
17	`⊆ = ⊇°`	`xs⊆ys⇔ys⊇xs`
18	`T°` $\frac{T}{\ni}$ `⊑ ∈`	every T-value of a point is a member of its T-set: `Λ(T)` is a map and `Λ(T)∋=T`.

(10c)



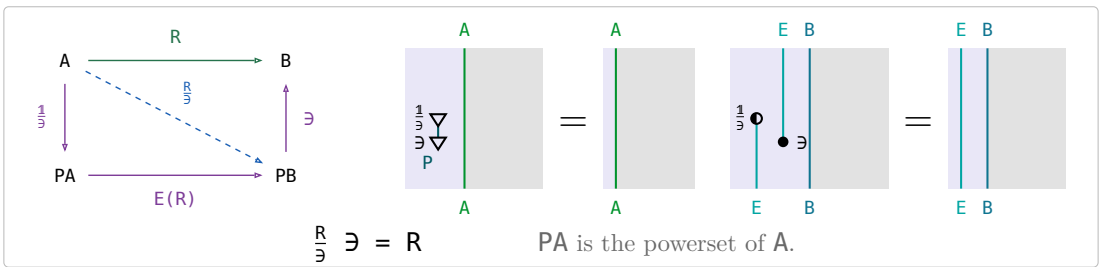
$\frac{R}{\ni}\ \ni = R$ $\quad$ Simple(F) ⇒ $F \sqsubseteq \frac{\mathbb{1}}{\ni}\ E(F\ \ni)$ $\quad$ Map(f) ⇒ $f\ \frac{\mathbb{1}}{\ni} = \frac{\mathbb{1}}{\ni}\ E(f)$



$\frac{R}{\ni} = \frac{\mathbb{1}}{\ni}\ E(R)$ $\quad$ $\frac{S}{\ni}\ E(R) = \frac{S\ R}{\ni}$

§10.1. i⊣E Power Allegory defined as adjunction

(10.1a)

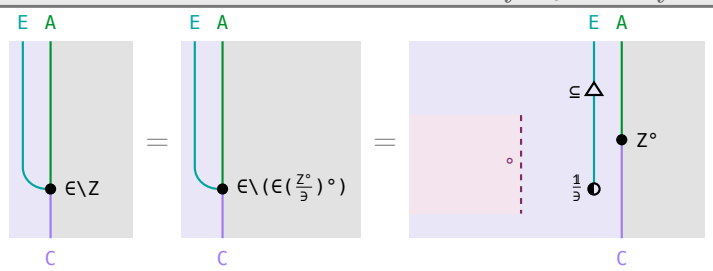


i is the inclusion Map(𝒜)→𝒜, doing nothing, so E is both the functor and the monad iE.

§10.2. ∈\ as a composite

$$\in \setminus Z = \subseteq (\tfrac{Z^\circ}{\ni})^\circ \qquad (10.2a)$$

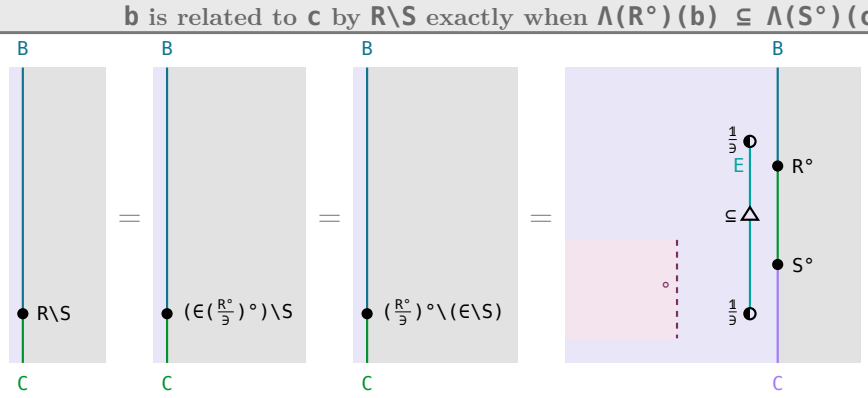
xs is related to c by ∈\Z exactly when xs ⊆ Λ(Z°)(c), the set of every a with a Z c



1	∈ \ Z	Z=∈Λ(Z°)° — $\frac{R}{\ni}$ ∋ = R
2	= ∈ \ (∈ ($\frac{Z^\circ}{\ni}$)°)	X\(Yf°)=(X\Y)f°, f a map
3	= ⊆ ($\frac{Z^\circ}{\ni}$)°	

$$R \setminus S = \tfrac{R^\circ}{\ni} \subseteq (\tfrac{S^\circ}{\ni})^\circ \qquad (10.2b)$$

b is related to c by R\S exactly when Λ(R°)(b) ⊆ Λ(S°)(c), the R-preimage of b inside the S-preimage of c



1	R \ S	R=∈Λ(R°)° — $\frac{R}{\ni}$ ∋ = R
2	= (∈ ($\frac{R^\circ}{\ni}$)°) \ S	(XY)\S=Y\(X\S)
3	= ($\frac{R^\circ}{\ni}$)° \ (∈ \ S)	f°\X=fX; ∈\S=⊆Λ(S°)° — ∈ \ Z = ⊆ ($\frac{Z^\circ}{\ni}$)°
4	= $\frac{R^\circ}{\ni}$ ⊆ ($\frac{S^\circ}{\ni}$)°	

§10.3. 𝒜≅Kleisli(E)

(10.3a)

$\mathsf{E(R)} \triangleq \frac{\ni R}{\ni}$, $\frac{\mathbb{1}}{\ni}$ $\mathsf{: A\to PA}$, $\mathsf{union = E(\ni)\ : PPA\to PA}$

$\mathsf{f \diamond g \triangleq f\ E(g)\ union}$, for $\mathsf{f : A\to PB}$ and $\mathsf{g : B\to PC}$

the monad is on $\mathsf{Map}(\mathcal{A})$, not on the allegory: E is a relator on all relations, but $\frac{\mathbb{1}}{\ni}$, union and $\mathsf{f\ E(g)\ union}$ are maps, and the Kleisli construction happens where they live.

(10.3b)

$$\boxed{\tfrac{S}{\ni} \diamond \tfrac{R}{\ni}} \overset{\text{definition of } \diamond,\ \mathsf{union=E(\ni)}}{=\!=\!=} \boxed{\tfrac{S}{\ni}\ \mathsf{E}(\tfrac{R}{\ni})\mathsf{E}(\ni)} \overset{\mathsf{E} \text{ a functor, } \mathsf{E(X)E(Y)=E(XY)}}{=\!=\!=} \boxed{\tfrac{S}{\ni}\ \mathsf{E}(\tfrac{R}{\ni}\ni)}$$

$$\overset{\frac{R}{\ni}\ \ni\ =\ R\ ,\ \frac{R}{\ni}\ni=R}{=\!=\!=} \boxed{\tfrac{S}{\ni}\ \mathsf{E(R)}} \overset{\frac{S}{\ni}\ \mathsf{E(R)}\ =\ \frac{S\ R}{\ni}\ ,\ \text{absorption}}{=\!=\!=} \boxed{\tfrac{SR}{\ni}}$$

the first three steps only unfold $\diamond$ — E is a functor, so $\mathsf{E}(\frac{R}{\ni})\mathsf{union}=\mathsf{E}(\frac{R}{\ni}\ni)=\mathsf{E(R)}$ and the union is gone. Absorption, $\frac{S}{\ni}\ \mathsf{E(R)}\ =\ \frac{S\ R}{\ni}$, is the whole law, and it is the functoriality of $\frac{\cdot}{\ni}$.

$\mathbb{1}$ goes to $\frac{\mathbb{1}}{\ni}$, the Kleisli identity, by definition — so $\frac{\cdot}{\ni}$ is an isomorphism of categories $\mathsf{𝒜≅Kleisli(E)}$, and §Section 10.1.'s $\mathsf{i⊣E}$ is the Kleisli adjunction. i is the identity on objects, so every object of the allegory is free.